Equitable Urban Electric Vehicle Charging: Feasibility and Benefits of Streetlight Charging in Kansas City Right-of-Way

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Abstract:

With an increasing global emphasis on sustainability, electric vehicles (EVs) play a crucial role in reducing urban pollution and carbon emissions. For EVs to be widely adopted, equitable and convenient access to charging infrastructure is essential. This research explores the potential of utilizing streetlights—ubiquitous elements of urban electrical infrastructure—as low-cost, equitable EV charging solutions. Compared to conventional chargers, streetlight charging offers several notable advantages, including proximity to roadways, potential boosts to the local economy, and easier usage due to city ownership. These chargers also leverage existing power setups to minimize costs and maximize efficiency by reutilizing existing structures. In this manuscript, we introduce a systematic framework to develop, analyze, and evaluate a scalable and cost-effective streetlight charging solution. Initially, we employ a two-tiered site selection framework that accounts for both charging demand and equity considerations to identify optimal locations for streetlight chargers. We then assess the feasibility of these chargers through observational data from 23 units installed in Kansas City, Missouri, comparing their performance with traditional chargers. Our evaluation extends to the environmental impact, comparing reduced gasoline consumption and greenhouse gas (GHG) emission reductions between streetlight and regular charging stations. Our findings reveal that streetlight charging stations offer faster charging speeds than conventional stations, validating their viability for EV charging. Additionally, the duration of stay at streetlight chargers tends to be shorter, likely due to parking constraints or associated costs. Furthermore, streetlight-powered chargers demonstrate considerable environmental advantages, achieving 11.94% greater avoided gasoline consumption and 11.24% higher GHG reductions compared to regular charging stations, highlighting their substantial environmental benefits.

Keywords: Electric Vehicle Charging Station, Equity, Demand Prediction, Streetlight Charging Station, GHG Emission Reduction

1. Introduction

With the growing global emphasis on sustainability and environmental protection, electric vehicles (EVs) have emerged as a key solution for reducing urban pollution and carbon emissions, while also contributing to a more diverse and resilient transportation fuel portfolio in the United States. A study by Kapustin & Grushevenko (2020) projects that, depending on the scenario, electric vehicles could account for 11-28% of the global road transport fleet by 2040. Addressing technical challenges, such as limited driving range and long recharging times, is crucial for increasing EV market penetration (Kchaou-Boujelben 2021). A key step in overcoming these challenges is the development and operation of extensive EV charging stations (Yuvaraj et al., 2024), which are critical for reducing "range anxiety" — the fear that a vehicle's battery will not have sufficient range to reach its destination, leaving its occupants stranded.

Ensuring equitable and convenient access to charging infrastructure is essential for supporting widespread EV adoption. Research indicates that low-income populations, communities of color, and residents of multiunit dwellings currently have the least access to charging infrastructure (Huether 2021; Werthmann & Kothari, 2021). Due to their lower fuel and maintenance costs compared to internal combustion engine vehicles, EVs—especially used ones—could be an affordable option for lower-income households. Unfortunately, despite significant investments like the $5 billion National Electric Vehicle Infrastructure (NEVI) Program to expand public charging infrastructure and develop alternative fuel corridors (AFCs), as well as state rebate programs for home garage EV chargers, many residents simply lack the privilege of owning a dedicated garage for convenient charging. This issue is particularly pronounced for renters and residents of multifamily housing (MFH), where garage or driveway charging is simply not an option. Without convenient and affordable access to charging, prospective EV buyers from these underserved communities and lower-income households are excluded from the advantages of electric vehicles.

In this research, we explore the feasibility of utilizing streetlights—existing electrical infrastructure commonly found throughout cities—as a means of providing low-cost and equitable EV charging solutions. Compared with conventional chargers, streetlight charging may offer several multifaceted benefits:

(1) *Proximity to Roadways and Contribution to the Local Economy*: Streetlights are typically located next to roads, often with adjacent curbside parking. This allows vehicles to conveniently park and charge while drivers engage in other activities such as shopping, dining, and running errands, which also boosts the local economy.

(2) *City Ownership that Eases Usage*: As municipal assets, streetlights fall under city control, facilitating their retrofitting and repurposing for EV charging without the complexities of private ownership.

(3) *Power Readiness for Reduced Costs*: Streetlights are already connected to the electrical grid, potentially allowing EVs to draw electricity directly from this infrastructure without the need for extensive rewiring.

(4) *Cost Efficiency by Reusing Existing Infrastructure*: Utilizing streetlights for EV charging can be a low-cost solution as it eliminates the need for laying conduit, digging trenches, cutting through curbs, or jackhammering sections of sidewalk.

In this research, we present a systematic framework to develop a cost-effective and scalable streetlight charging solution, analyze its feasibility, and evaluate its performance. This paper provides valuable insights for researchers, practitioners, and communities interested in deploying streetlight charging stations. Specifically, the contributions of this study are threefold:

(1) *Two-Tiered Site Selection Framework*: This study proposes a two-tiered site selection framework to identify optimal candidate sites for streetlight chargers. We consider both charging demand through a boosting-based ensemble learning (BEL) model and equity implications through an EV adoption opportunity-based siting model.

(2) *Feasibility Analysis*: After identifying and installing a total of 23 streetlight chargers in Kansas City, Missouri, with support from a Department of Energy (DOE) grant, we test the performance of these streetlight charging stations and compare their metrics with those of regular chargers to assess their feasibility. We aim to determine whether these new streetlight-powered chargers can charge EVs at speeds that are comparable to or even exceed those of regular chargers.

(3) *Benefit Analysis*: We showcase the benefits of streetlight chargers by comparing reduced gasoline consumption and greenhouse gas (GHG) emission reductions between streetlight and regular charging stations. This analysis provides insights into new pathways for saving energy, reducing carbon emissions, and decreasing air pollution in urban development.

The overall research framework supporting the development of a streetlight charging network is illustrated in Figure 1. Our methodology begins with a comprehensive data collection phase (depicted in the middle box), which gathers a variety of data sources including: charging logs from both regular and streetlight chargers; detailed information on streetlights such as location, nearby parking availability, and power voltage; land use types and density; historical weather data; traffic patterns surrounding the area; points of interest via the Google Maps API; and EV registration details. For a more detailed description of these data sources, please see Sections 3.1 and 3.2.

Building on this rich dataset, we employ a two-tiered site selection framework (shown in the left box) to identify optimal locations for installing new chargers. Post-installation, we monitor the usage data of these chargers over a 12-month period, from January to December 2023. This phase is followed by feasibility testing (highlighted in the top right box) and a benefit analysis (indicated in the lower right box) to assess the performance and benefits of the newly installed chargers.

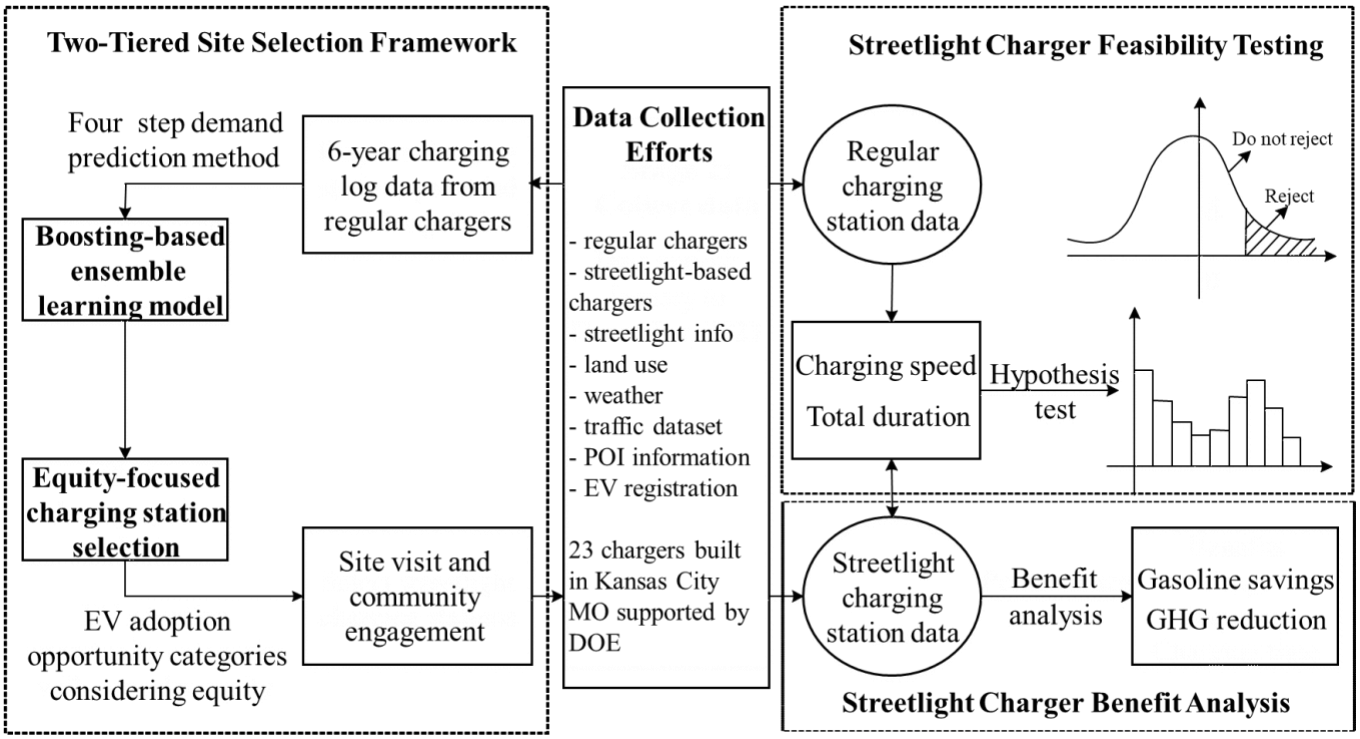


Figure 1 Streetlight charging research framework

The remainder of this paper is organized as follows. Section 2 reviews related work on the EV charging station location problem and performance analysis. Section 3 proposes a two-tiered site selection framework for the transformation of streetlight charging stations, considering both demand and equity implications. Section 4 presents the methodology and results of feasibility performance testing. Section 5 analyzes the benefits of the constructed streetlight charging stations in Kansas City, Missouri. Finally, Section 6 presents the conclusions.

2. Literature review

The optimal planning of charging infrastructure, known as the charging station location problem (CSLP), involves selecting the best locations for charging stations to maximize the fulfillment of EV drivers' recharging demands within the constraints of an investment budget (Kchaou-Boujelben 2021). Research on CSLP can be categorized into flow-based and node-based approaches. Flow-based models (Capar et al., 2013; Yıldız et al., 2016) typically represent charging demand as a series of origin-destination (O-D) trips. In this approach, DC fast charging stations are strategically placed to ensure that the distance between consecutive stations is within the driving range of EVs, allowing drivers to complete their trips without depleting their batteries. On the other hand, node-based models (Cavadas et al., 2015; Huang et al., 2016; Cui et al., 2019) assume that charging demand in a road network is independent of specific EV trips and occurs at various nodes throughout the network. In this approach, node-based demand typically arises from

groups of EV drivers who prefer to recharge their vehicles at stations conveniently located near their homes, workplaces, or other points of interest (POIs) such as plazas. The charging demand is usually represented by the size of the charging station's service area. Yi et al. (2021) forecasts monthly commercial EV charging demand using a deep learning Sequence-to-Sequence model, validated with real-world data from the State of Utah and the City of Los Angeles.

Various studies have modeled the optimization of EV charging stations for infrastructure planning. For instance, Zhang et al. (2016) proposed a comprehensive planning framework within urban areas aimed at minimizing the social cost of the entire electric vehicle charging system. He et al. (2015) examined the optimal location of public EV charging stations in a road network, considering driver travel and charging decisions to minimize travel and charging time while ensuring sufficient power for trips. Sun et al. (2016) investigated how EV users choose fast charging stations and the extent of detours they are willing to make. He et al. (2016) focused on site selection planning in Beijing, incorporating local supply and demand constraints into the facility location model. However, these studies lack consideration of social equity in EV charging planning. Luo et al. (2018) proposed planning for EV charging centers with various facilities to minimize the annual social cost of the entire system. Klein et al. (2020) used agent-based simulation to study the purchasing behavior of German consumers for electric and plug-in hybrid vehicles. Chen et al. (2020) analyzed EV charging behavior patterns using functional data analysis. Recently, Song & Hu (2023a) proposed a learning-based supply-demand coupling optimization model, integrating charging demand management into infrastructure planning with a two-stage approach for site selection. Despite these advances, these studies do not consider retrofitting existing infrastructure into charging stations to reduce costs and time. Furthermore, multiple studies have evaluated the usage and performance of existing EV charging infrastructure. Robinson et al. (2013) analyzed the Switch EV trial in Northeast England, focusing on various use cases and their impact on carbon dioxide emissions. Sun et al. (2015) proposed using a mixed logit model to study the choice behavior of pure electric vehicle users regarding charging time. Mills & Macgill (2017) studied the greenhouse gas emissions of plug-in hybrid electric vehicles under a range of standard electric vehicle powertrain integration scenarios.

In summary, existing CSLP research often focuses on demand considerations and using predicted demand to build new charging infrastructure with the aim of maximizing charger usage or minimizing charging time. However, these studies often overlook critical aspects such as social equity in site selection and the potential benefits of retrofitting existing infrastructure, like streetlights, to reduce costs and implementation time. Furthermore, previous research on the performance and environmental impact of EV infrastructure can inform the retrofitting of existing infrastructure for EV charging.

3. Streetlight charger site selection framework

In this section, we introduce a two-tiered site selection model (as shown in the left box of Figure 1) designed to identify optimal public streetlights for EV charging. This model integrates two critical components: a four step charging demand prediction method leveraging a boosting-based ensemble learning model, and

an equity-focused siting selection approach based on EV adoption opportunities. Detailed explanations of both components and how they interact to guide the selection process will be discussed in detail in the remainder of this section. Then, site visit and community engagement are introduced to finalize the site selection.

3.1 Four step demand prediction method

The demand prediction model identifies candidate locations for streetlight charging using a four-step approach, detailed in Algorithm 1 below. It utilizes six years of data from 455 charging stations, travel demand data, and other sources in Kansas City, Missouri to extract influencing factors and the charging station usage frequency (Step 1). Key factors include land use, station density, traffic volume, and trip production. A boosting-based ensemble learning (BEL) model is then trained to predict the weekly usage frequency based on those defined factors (Step 2). Potential high-usage locations, including apartments, shopping malls, and restaurants, are identified for deploying charging stations (Step 3). Finally, the weekly usage frequency of 1,252 candidate locations is evaluated, and 300 streetlights with the highest predicted usage are selected for further survey (Step 4).

Algorithm 1 The four step demand prediction method

Step 1: Define influencing factors

Identify factors affecting Electric Vehicle Supply Equipment (EVSE) usage frequency, such as existing charging station density, traffic volume, trip production, and land use types.

Step 2: Develop a boosting-based ensemble learning model

Train a boosting-based ensemble learning model using weekly usage frequency of existing charging stations as the dependent variable and the factors from Step 1 as independent variables.

Step 3: Identify candidate locations

Use the Google Maps API to retrieve POIs in Kansas City as candidate locations.

Step 4: Predict usage frequency of candidate locations

Select 300 streetlights at locations with the highest predicted usage rates for further field survey.

3.1.1 Feature definition and model development

In step 1, the selection of these 15 features is based on a comprehensive consideration of EV charging demand: (1) Spatial Context Information (e.g., charging station density and land use types) reflects the usage characteristics of the area where the charging station is located. For example, areas with high charging station density may experience increased competition, while different land use types (such as commercial or residential areas) can affect user charging behavior and frequency. (2) Weather Information (e.g., precipitation, temperature, and wind speed) is closely related to vehicle usage frequency. Adverse weather conditions may reduce travel, thereby decreasing charging demand, while extreme temperatures can affect battery performance, leading to more frequent charging needs. (3) Charger Type (DC fast charger and Level 2 charger) directly influences charging speed and user choice. Fast chargers typically attract users seeking

short charging durations, while slower chargers are more suitable for long-term parking. (4) Traffic Information (e.g., annual average daily traffic and trip production) measures the vehicle flow and potential charging demand in the area surrounding the charging station. Areas with high traffic volume usually correlate with higher charging demand. A detailed definition is provided in Table 1, with reference to Pan et al. (2025).

Table 1 Definition of 15 features and response variable

Category	Notation	Definition and unit	Data type
Response Variable	y_{ij}	Number of charging events at station i during week j	Numerical
Spatial Context Information	D_i	Charging station density in the same zip code as station i (#charging stations/sq mile)	Numerical
	$LU1_i$	Land use type: station i located in an institutional area	Binary
	$LU2_i$	Land use type: station i located in a transportation area	
	$LU3_i$	Land use type: station i located in a commercial area	
	$LU4_i$	Land use type: station i located in a residential area	
	$LU5_i$	Land use type: station i located in a recreational area	
	$LU6_i$	Land use type: station i located in a vacant area	
	$LU7_i$	Land use type: station i located in an industrial area	
Weather Information	P_j	Total precipitation during week j (mm)	Numerical
	T_j	Average temperature during week j $(°C)$	
	W_j	Average wind speed during week j (m/sec)	
Charger Type	$C1_i$	Availability of a DC fast charger at station i	Binary
	$C2_i$	Availability of a Level 2 charger at station i	
Traffic Information	$AADT_i$	Annual average daily traffic near station i	Numerical
	TP_i	Trip production in the traffic analysis zone of station i	

With these 15 features defined, $\mathbf{x}_{ij}$, representing the array of features for charging station i during week j, can be expressed using Eq. (1) below.

$$\mathbf{x}_{ij} = \left[D_i, P_j, T_j, W_j, C1{\sim}2_i, LU1{\sim}7_i, AADT_i, TP_i\right] \quad (1)$$

Then, with these 15 features and the target variable established, the step 2 of training the BEL model can be then implemented. Ensemble learning models can enhance prediction accuracy by training multiple sub-models on the same data and aggregating their outputs (Dong et al., 2020; Song & Hu, 2023b). Boosting improves weak learners by iteratively adjusting the data weights based on prior errors, and the final model

computes a weighted sum of these predictions. The objective function $\mathcal{J}(\mathbf{F})$ is defined in Eq. (2), where the goal is to minimize the loss function ℓ along with a regularization term $\mathcal{R}$ to prevent overfitting:

$$\min_{\mathbf{F}} \mathcal{J}(\mathbf{F}) = \sum_{i=1}^{N} \sum_{j=1}^{W} \ell\left(\hat{y}_{ij}, y_{ij}\right) + \sum_{k=1}^{K} \mathcal{R}(h_k) \tag{2}$$

Here, $\mathbf{F}$ is the set of K sub-models as Eq. (3), where each h_k represents a regression tree learning distinct characteristics from the dataset.

$$\mathbf{F} = \{h_1, h_2, \ldots, h_K\} \tag{3}$$

The loss function ℓ is crucial in the charging demand prediction model. It is defined as the squared error between the predicted charging demand $\hat{y}_{ij}$ and the actual value y_{ij} from field data, as shown in Eq. (4). The model's performance is assessed by minimizing this difference, which helps reduce bias and enhance prediction accuracy.

$$\ell\left(\hat{y}_{ij}, y_{ij}\right) = \left(y_{ij} - \hat{y}_{ij}\right)^2 \tag{4}$$

The regularization term $\mathcal{R}(h_k)$ is formulated in Eq. (5) to control model complexity. $|L|$ is the number of leaf nodes in tree h_k, α controls complexity, w_ℓ denotes the weight of each leaf node ℓ, and β adjusts the regularization strength.

$$\mathcal{R}(h_k) = \alpha|L| + \frac{\beta}{2}\sum_{\ell \in L} w_\ell^2 \tag{5}$$

The final predicted value $\hat{y}_{ij}\left(\mathbf{x}_{ij}\right)$ for a given input vector $\mathbf{x}_{ij}$ is obtained by summing the outputs of all sub-models, as shown in Eq. (6).

$$\hat{y}_{ij}\left(\mathbf{x}_{ij}\right) = \sum_{k=1}^{K} h_k\left(\mathbf{x}_{ij}\right) \tag{6}$$

With an adequate number of sub-models, the BEL model is well-trained and can generate reliable charging demand forecasts for various charging station locations.

3.1.2 Model training results

The prediction performance of the BEL model on the training and testing dataset is measured by R-square, root mean squared residuals (RMSE) and mean absolute error (MAE). As shown in Figure 2, R-square values are 0.72 and 0.63, RMSE values are 2.38 and 2.77, and MAE values are 1.62 and 1.85 on the training and testing subsets, respectively. Compared with the charging demand prediction models from the literature (Almaghrebi et al., 2020), whose training and testing R-square values were shown to be between 0.567 and

0.519, the proposed model demonstrates satisfactory performance in generating accurate charging demand predictions.

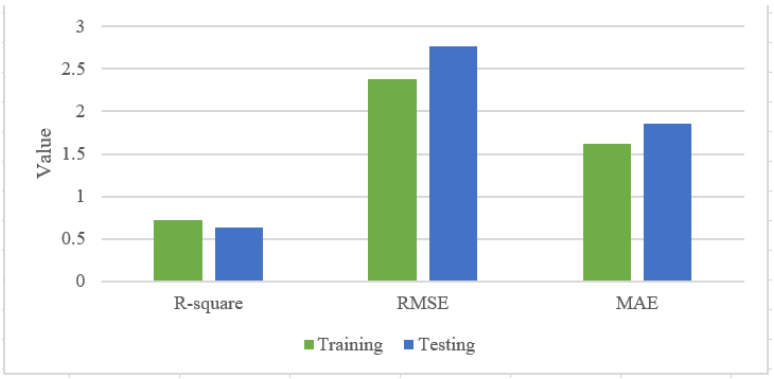


Figure 2 Prediction performance on training and testing sets

To further validate the prediction performance of the BEL model, three other representative prediction models are implemented using the same dataset for comparison: Multiple Linear Regression (MLR), K-Nearest Neighbors (KNN), and Neural Networks (NN). Figure 3 shows that the proposed model outperforms the other prediction models with the highest R-square value and the lowest RMSE and MAE. Thus, the BEL model provides relatively accurate prediction results due to its ability to utilize boosting techniques to combine weak learners into a strong predictive model.

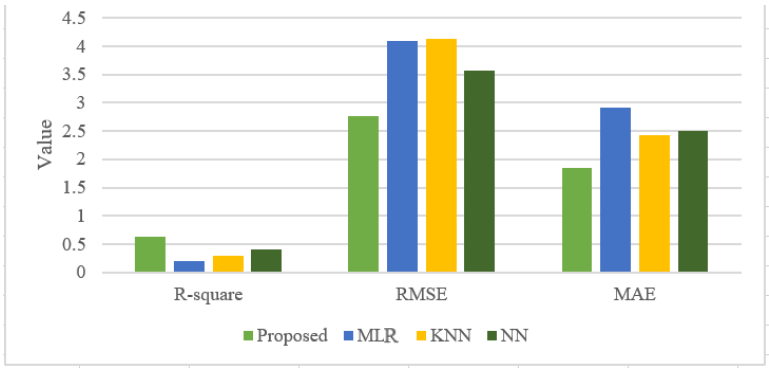


Figure 3 Performance comparison among four models

3.2 Equity-focused site selection model

This section presents an equity-focused model for identifying streetlight charging locations in residential areas to ensure fair investment in Kansas City's EV infrastructure. The model targets geographic areas that serve low-income and vulnerable populations, focusing on individuals without access to private residential charging. By integrating demographic criteria with EV registration data, land use patterns, Kansas City parcel data, and environmental indicators from the EPA's EJScreen tool, the model aims to balance equitable access with market demand. This approach ensures that publicly funded infrastructure is

distributed fairly, addressing both current needs and potential future adoption. Key considerations for the model are outlined as follows:

1. *Focus Area 1 – Easy Wins:* These are areas with high EV shares and poor residential EVSE availability. The equity consideration is whether we can support residential charging for existing EV owners.
2. *Focus Area 2 – Unlock Potential:* These areas have low EV shares, demographics supportive of EV adoption, and poor residential EVSE availability. The equity consideration is whether we can enhance residential charging to encourage EV adoption in these areas.
3. *Focus Area 3 – Create Opportunity:* These areas have low incomes and high multi-family housing shares, indicating poor residential EVSE access. The equity consideration is whether we can target areas with vulnerable populations lacking EVSE access.

To support the equity-focused charging station selection model, the data layers listed in Table 2 are utilized. Combined with the BEL model, these layers are overlaid onto the BEL output to identify locations in Kansas City that facilitate a robust charging network, address the residential charging needs of current and future EV drivers.

Table 2 Definition of data layers in the equity-focused model

Data Layers	**Description**
(1) EV and Plug-in Hybrid Electric Vehicle (PHEV) data from 2014-2017 (IHS Market data)	a) Weight PHEVs higher than EVs, as it is likely PHEVs will use Level 2 chargers more
	b) Weight ZIP codes where growth is faster over time (implying faster adoption)
(2) Multi-unit dwelling/land use (Kansas City, MO, data)	Building type of individual buildings in Kansas City
(3) Rental data	Block and block group housing data in Jackson County, Missouri
(4) Sociodemographic indicators for "likely to adopt" groups	Demographic data that align with EV ownership (e.g., income, family type, homeownership, age, education)
(5) EJScreen layers	a) Environmental indicators i. Particulate matter (PM2.5) ii. National Air Toxics Assessment (NATA) cancer risk iii. Lead paint indicator iv. Hazardous waste proximity v. Ozone vi. NATA respiratory hazard index (HI) vii. Superfund proximity viii. Wastewater discharge indicator

	ix. NATA diesel PM x. Traffic proximity xi. Risk management plan (RMP) proximity
	b) Demographic indicators i. Minority population ii. Low-income population iii. Linguistically isolated iv. Less than high school education v. Under age 6 vi. Over age 64

The details of the proposed equity-focused charging station selection model are shown in Algorithm 2.

Algorithm 2 Equity-focused charging station selection model

Step 1: Input data collection

Step 1.1: Collect EV and PHEV data (2014-2017)

Weight PHEVs higher due to higher likelihood of using Level 2 chargers and weight ZIP codes with faster growth rates.

Step 1.2: Gather multi-unit dwelling and land use data

Identify building types in Kansas City, MO.

Step 1.3: Obtain rental data

Collect block and block group housing data in Jackson County, Missouri.

Step 1.4: Compile sociodemographic indicators

Align data with likely EV ownership demographics (income, family type, homeownership, age, education).

Step 1.5: Fetch EJScreen layers

Include environmental indicators (e.g., PM2.5, cancer risk, lead paint, hazardous waste proximity) and demographic indicators (e.g., minority population, low-income population, linguistic isolation).

Step 2: Demographic analysis

Identify geographic areas meeting specific criteria from Table 2.

Step 3: Heat map creation for each focus area

Generate three separate heat maps based on the data layers described in Table 4. Overlay demographic, EV registration, and land use data to visualize potential locations.

Step 4: Integrate with BEL model output

Overlay the equity- Equity model heat maps onto the BEL model output. Identify high-utilization streetlight charging locations supporting a robust network.

Step 5: Equity consideration

Focus on areas supporting residential charging for existing EV owners. Enhance residential charging to encourage adoption in areas with supportive demographics.

Step 6: Final charging location identification

Use combined data and models to determine final streetlight charging locations. Ensure locations meet residential charging needs and are expected to have high utilization.

Step 7: Generate list of charging locations

Provide a list of recommended streetlights charging locations. Include detailed information on each location's support for equitable EVSE access.

3.3 Site visits and community engagement

Using a two-tiered site selection model, this study identifies the top candidate sites in Kansas City for streetlight chargers. The evaluation process combines virtual assessments—using GIS and mapping tools—with in-person visits to verify each site's feasibility. Key factors include the type of streetlight poles and parking conditions. Poles that are decorative, historic, or require extensive modifications are excluded. Parking restrictions also narrow the options by eliminating locations near fire hydrants, driveways, intersections, bike lanes, or other areas lacking safe, legal parking. In-person visits help uncover pedestrian hazards and other practical challenges not visible on maps. Cost is a primary consideration, with priority given to sites requiring minimal additional expenses, such as those near transformers or without pavement removal.

Following the site visits, the list is refined to approximately 40 potential locations. Then, community engagement events are held to gather feedback from local stakeholders, ensuring the selected sites align with community needs and preferences. After incorporating input on additional factors—such as cell phone reception, charger orientation, traffic safety, and nearby amenities—23 final candidate sites are chosen (see Figure 4). Installation costs vary between $13,000 and $45,000 per site, depending on the scope of required electrical upgrades.

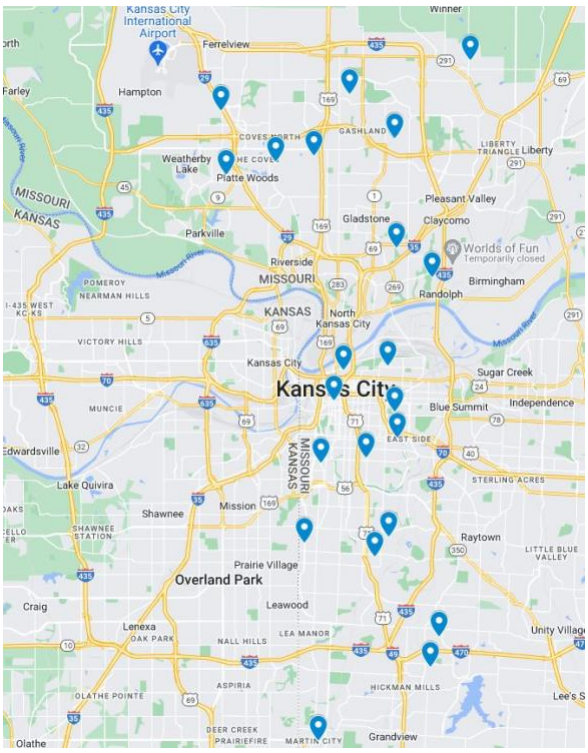


Figure 4. Final 23 sites for streetlight chargers

4. Feasibility testing of streetlight chargers

In this section, we test the performance of the streetlight charging stations (illustrated in the top right box of Figure 1) in terms of charging speed and duration of charging events, and then compare these metrics with those of regular chargers to demonstrate their feasibility. The hope is that these new streetlight-powered chargers can charge EVs at a speed comparable to that of regular chargers. To achieve this, we develop and apply a Mann-Whitney U test-based hypothesis testing model using real-world data collected from the 23 selected streetlight-powered chargers.

4.1 Hypothesis test methodology

We employ the Mann-Whitney U test to examine the medians of the distribution data, comparing the performance between regular charging stations and streetlight-powered charging stations. This provides a theoretical foundation for further analysis of the feasibility of retrofitting streetlights as charging stations.

- **Null hypothesis H_0**: The median performance of the streetlight charging station is not different from that of a regular charging station, i.e., $m_1 = m_2$.
- **Alternative hypothesis H_1**: The median performance of the streetlight charging station is different from that of a regular charging station, i.e., $m_1 \neq m_2$.

1 For the algorithm used to test the median performance, see Algorithm 3.

Algorithm 3 Mann-Whitney U test-based hypothesis testing

Step 1: State the hypotheses

Step 2: Combine the samples

Combine the two samples into a single dataset and rank all the values. If there are tied values, assign the average rank.

Step 3: Calculate the rank sums

Calculate the sum of ranks for each sample:

R_1= sum of ranks for the streetlight charging station

R_2= sum of ranks for the regular charging station

Step 4: Calculate the test statistic

The test statistics U_1 and U_2, and for the two samples are calculated as follows:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

The test statistic U is the smaller of the two values:

$$U = \min(U_1, U_2)$$

Step 5: Determine the critical value, p-value and Z-score

The null hypothesis in Mann-Whitney U testing can be rejected by either using critical values, or p-values. To determine these two values, the Mann-Whitney U distribution table or relevant software can be used. As illustrated in Figure 5 below, if the significance level α is set at 0.05, corresponding to a 95% confidence level, the critical value is identified as 1.96.

Next, we calculate the z-score. For large samples (typically n≥20), the distribution of U can be approximated by a normal distribution with mean μU and standard deviation σU:

$$\mu U = \frac{n_1 n_2}{2}$$

$$\sigma U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

Then, the z-score can be calculated as:

$$z = \frac{U - \mu U}{\sigma U}$$

Step 6: Make a decision

To determine whether to reject the null hypothesis in a Mann-Whitney U test, consider the following criteria:

1. Compare the test statistic to the critical value: Calculate the test statistic (U or the z-score) and compare it to the critical value, which is 1.96 if a 95% confidence level is adopted. If the test statistic exceeds the critical value, reject the null hypothesis.
2. Compare the P-value with the significance level (α): If the p-value is less than α, reject the null hypothesis. For instance, with α set at 0.05, any p-value less than 0.05 indicates a significant result, prompting rejection of the null hypothesis.
3. Conclusion if neither criterion is met: If neither of the above conditions is satisfied—that is, if the test statistic does not exceed the critical value and the p-value is not less than α—then do not reject the null hypothesis.

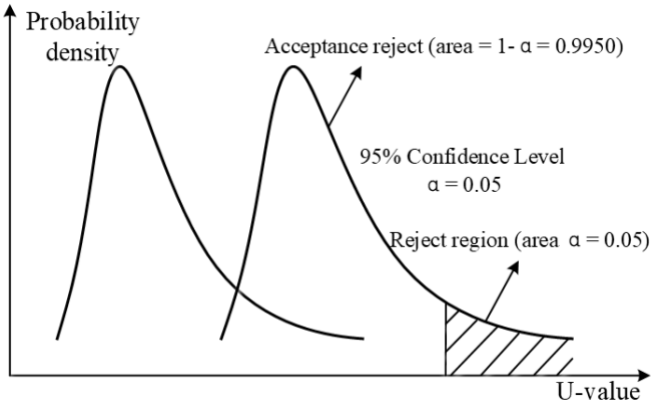


Figure 5 Mann-Whitney U test for equality of two mean

4.2 Feasibility analysis

Following the site selection framework discussed in Section 3, 23 streetlights were selected and converted into charging stations in December 2022 with funding support from the DOE. The research team monitored the charger usage continuously for a year, collecting and comparing charging log events from these streetlight-powered chargers with those from regular chargers. Below, we present the performance of these streetlight-powered chargers, in terms of charging speed, and charging duration, respectively.

4.2.1 Charging speed per charging event

We use Eq. (7) to determine the charging speed of the charging stations:

$$v_t = \frac{E_t}{T_t} \tag{7}$$

where v_t is the charging speed of the t time, E_t is the energy charged, T_t is the charging duration.

As illustrated in Figure 6, we first use violin plots to compare the charging speed distributions between streetlight charging stations (on the left) and regular charging stations (on the right). The X-axis displays

the two charging station types, while the Y-axis represents charging speed (kW). The black horizontal lines in each plot denote the median and quartile ranges. It can be observed that streetlight charging stations exhibit speeds primarily concentrated between 4 kW and 6 kW, with some lower values near 0 kW and higher speeds reaching 8 kW. The regular charging stations show a similar distribution but with a slightly wider range, reflecting more extreme speed values.

Then, we use the Mann-Whitney U test to test the hypothesis. The null hypothesis H_0 posits that there is no significant difference in charging speed between streetlight charging stations and regular charging stations. The alternative hypothesis H_1, on the other hand, asserts that there is a significant difference in charging speeds between the two types of stations. The result shows that the calculated z-value is 11.63, and the associated p-value is 0.027. When this p-value is compared with a 95% confidence level ($\alpha = 0.05$), both criteria for rejecting the null hypothesis are satisfied: the z-value of 11.63 exceeds the critical value of 1.96, and the p-value of 0.027 is below 0.05. Therefore, we can conclude that the null hypothesis should be rejected. This indicates a significant difference in charging speed between streetlight charging stations and regular charging stations. Further analysis reveals that the median charging speed of streetlight stations is 6.53 kW, which is significantly higher than the median of 6.00 kW at regular stations. Therefore, we can assert that streetlight charging stations offer significantly faster charging speeds compared to regular charging stations.

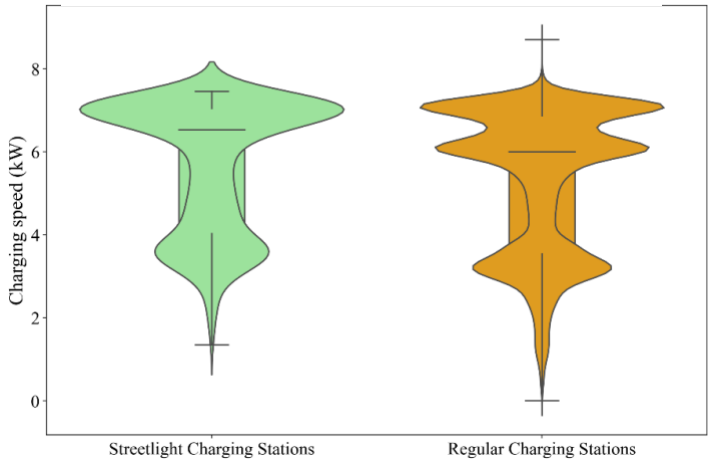


Figure 6 Comparison of charging speed distribution per event

4.2.2 Total duration per charging event

Figure 7 compares the total charging duration per event, illustrating the distribution of charging times for streetlight charging stations and regular charging stations. We can observe that the charging durations for streetlight stations are primarily concentrated between 0 and 5 hours, indicating a more compact distribution, while regular stations exhibit a broader range, with some durations extending up to 25 hours. Therefore, the null hypothesis H_0 is that the charging duration of the streetlight charging station is not significantly

different from that of a regular charging station. The alternative hypothesis H_1 is that the charging duration of the streetlight charging station is significantly different from that of a regular charging station.

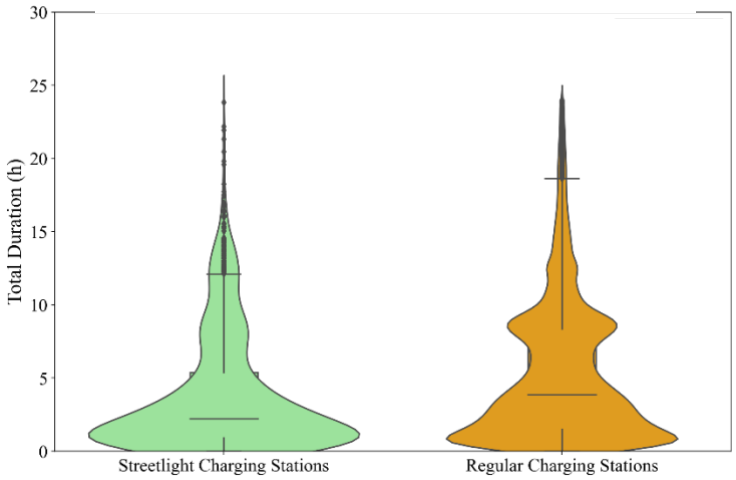


Figure 7 Comparison of total duration distribution per event

The result of the Mann-Whitney U test shows that the z-value is 14.85, with an associated p-value of 0.067. Although the p-value does not meet the 95% confidence level criterion (α=0.05), the criterion based on the z-value is satisfied, as 14.85 significantly exceeds the critical value of 1.96. Hence, we can conclude that the null hypothesis should be rejected based on the z-value criterion. Further analysis reveals that the median duration at streetlight stations is 2.23 hours, which is considerably shorter than the 3.84 hours at regular stations. This suggests that EV drivers typically spend less time at streetlight charging stations than at regular ones. This reduced duration may be due to factors such as parking constraints, the costs associated with on-street parking, or the faster charging capabilities of streetlight stations.

5. Environmental benefit analysis of streetlight charging

The widespread adoption of EVs is essential for reducing fossil fuel use and greenhouse gas emissions. This section evaluates how integrating charging facilities into streetlights can amplify environmental benefits. By comparing streetlight charging stations with regular ones in terms of avoided gasoline consumption and GHG emission reductions, we highlight the advantages of this infrastructure. Our goal is to demonstrate how streetlight charging can advance environmental sustainability, as illustrated by the lower right box of Figure 1.

5.1 Avoided gasoline consumption

The avoided gasoline consumption represents the amount of gasoline that would have been used if internal combustion engine vehicles (ICEVs) were driven instead of EVs. We use Eq. (8) to calculate the gasoline saving of the charging stations.

$$AvoidedGas(t) = Fuel_{ICEV}(t) - Fuel_{EV}(t) \quad (8)$$

where $AvoidedGas(t)$ is the avoided gasoline consumption during time t, and $Fuel_{ICEV}(t)$ is the amount of gasoline an ICEV would have used over the same distance.

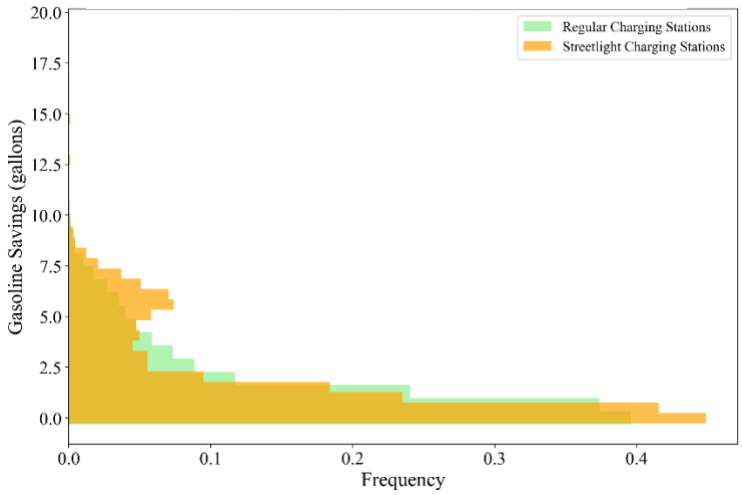


Figure 8 Comparison of avoided gasoline consumption distribution

Figure 8 compares avoided gasoline consumption between streetlight charging stations and regular stations, with most values falling between 0 and 5 gallons. However, streetlight stations show a wider spread, particularly in the 5 to 10-gallon range, suggesting they support longer or more frequent EV use, displacing more ICEV miles. On average, streetlight stations avoid 2.25 gallons per session—11.94% more than the 2.01 gallons from regular stations—highlighting their greater effectiveness in reducing gasoline consumption. These findings underscore the enhanced impact of streetlight charging stations in reducing energy consumption.

5.2 GHG emission reduction

Reducing GHG emissions is a primary goal of increasing EV adoption. We calculate the GHG emission reduction by considering the emissions that would have been produced by an ICEV, using Eq. (9).

$$GHGReduction(t) = AvoidedGas(t) \times EmissionFactor \quad (9)$$

where $GHGReduction(t)$ is the reduction in GHG emissions during time t, $AvoidedGas(t)$ is from Eq. (8), and $Emission\ Factor$ is the amount of GHG emissions per unit of gasoline consumed.

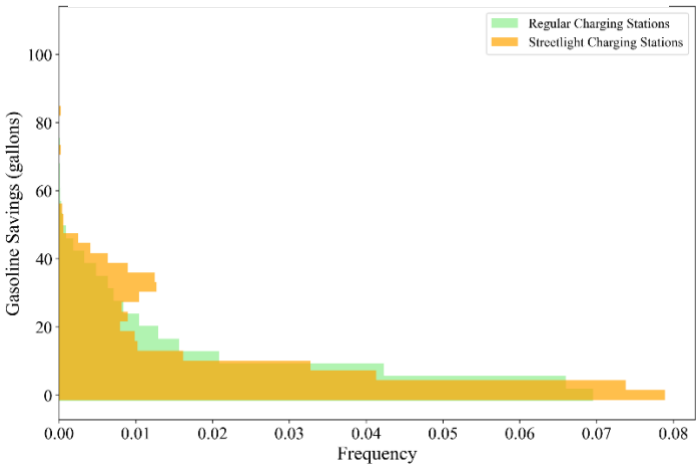


Figure 9 Comparison of GHG emission reduction distribution

Figure 9 compares the distribution of GHG emission reductions between streetlight and regular charging stations, showing higher frequencies in the 20–40 kg range and a broader overall spread for streetlight stations. This suggests that streetlight charging facilitates greater GHG reductions by enabling more substantial gasoline avoidance. On average, streetlight stations reduce emissions by 12.77 kg per session—11.24% more than the 11.48 kg for regular stations—further highlighting their environmental advantages.

By integrating charging infrastructure into streetlights, cities can promote greater EV usage, leading to significant reductions in gasoline consumption and GHG emissions. This strategy leverages existing urban infrastructure to support sustainable transportation and contributes to broader environmental goals.

6. Conclusions

In this study, we present a systematic framework to develop a cost-effective and scalable streetlight charging solution, analyze its feasibility, and evaluate its performance. The goal is to utilize existing urban infrastructure to improve equitable EV charging access, particularly for renters and multifamily housing residents. We introduce a two-tiered site selection approach, which includes a demand prediction model and an equity-focused station location selection model. Performance comparisons between streetlight and conventional chargers, supported by hypothesis testing, reveal that streetlight chargers offer faster charging speeds. A benefit analysis further underscores their environmental advantages, showing approximately 11.94% more avoided gasoline consumption and 11.24% greater GHG reductions. These findings validate the feasibility of streetlight chargers and highlight their potential to enhance urban sustainability by leveraging existing infrastructure.

7. Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

8. Acknowledgements

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9. Disclaimer

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